SERVICE REQUEST AND WORKFLOW AUTOMATION SYSTEM

With Student–Admin Integrated Management Platform

[1]D.Durga Prasad, [2] P.Kasi, [3]Y.V.D.Sai, [4]K.Sai Ram, [5]S.S.R.CH.PHANI KUMAR, [6] Mrs.B.REVATHI
[1]Student, [2]Student, [3]Student, [4]Student, [5]Student, [6]Professor
[1]Computer Science and Engineering,
[1]BVCEC, Odalarevu, India

***Abstract*:** In modern organizations, managing service requests manually through emails, spreadsheets, or paper-based forms often leads to delays, lack of transparency, and inefficient tracking. To overcome these challenges, this project presents a Service Request and Workflow Automation System developed using Java Full Stack technologies.

The system provides a centralized web-based platform where users can submit service requests digitally. Each request is automatically routed through predefined workflow stages such as validation, approval, task assignment, processing, and closure. Role-based access control ensures that users, administrators, and service staff have clearly defined permissions.

The Admin module allows user management, workflow configuration, task assignment, monitoring of request progress, and generation of reports. The User module enables request submission, status tracking, and real-time notifications.

The proposed system improves efficiency, reduces manual intervention, enhances transparency, and ensures structured workflow automation. The system is scalable, secure, and suitable for educational institutions, corporate offices, and enterprise environments.

***IndexTerms* – Service Request Management, Workflow Automation, Role-Based Access, Java Spring Boot, Web Application**

1. INTRODUCTION

In the digital era, organizations rely heavily on information systems to manage operational activities efficiently. One of the most common activities across organizations is handling service requests such as IT support, maintenance issues, approvals, and administrative services.
Traditional service request handling methods involve:

- Emails
- Phone calls
- Paper forms
- Manual tracking

These methods lack structured workflows and transparency, often leading to delays, lost requests, and miscommunication.
The proposed **Service Request and Workflow Automation System** provides a centralized digital platform where service requests are:

- Submitted online
- Assigned unique request IDs
- Routed automatically through predefined workflow stages
- Monitored in real-time

The system ensures automation, accountability, and efficient service management.

2. NEED FOR THE STUDY

The need for a Service Request and Workflow Automation System arises due to the following limitations in existing systems:

- Manual processing increases workload and human errors
- Lack of centralized tracking
- No predefined workflow stages
- Delayed approvals and task assignments
- Poor communication between departments
- Absence of real-time monitoring

3. LITERATURE SURVEY

Several systems have been developed for ticket management and service request tracking. Traditional systems include:

- Manual request registers
- Email-based ticketing
- Basic computerized ticket tools

Although these systems reduced paperwork, they lacked:

- Workflow automation
- Real-time notifications
- Role-based access control
- Analytics dashboards

Recent studies highlight the importance of workflow automation in improving efficiency and accountability. Modern frameworks like Spring Boot enable scalable and secure backend systems for enterprise-grade applications.
This project addresses the gap by implementing a complete automated workflow system using Java Full Stack technologies.

4. PROPOSED SYSTEM

The proposed system is a web-based application that automates the complete lifecycle of service requests.
Key Features:

- Online request submission
- Unique request ID generation
- Automated workflow routing
- Role-based authentication
- Task assignment and tracking
- Real-time notifications
- Reporting and analytics

Roles:

1. **User**
    - Submit service requests
    - Track request status
    - Receive notifications
2. **Admin**
    - Manage users
    - Configure workflows
    - Assign tasks
    - Monitor request progress
    - Generate reports

The system ensures structured and consistent processing of requests.

5. ARCHITECTURE OVERVIEW

The system follows a **three-tier architecture**:
1. Presentation Layer

- Developed using HTML, CSS, JavaScript, React
- Provides user interface for request submission and dashboards

2. Application Layer

- Java & Spring Boot
- Handles authentication, workflow logic, task routing

3. Database Layer

- MySQL
- Stores users, requests, workflow stages, logs

REST APIs enable communication between frontend and backend.

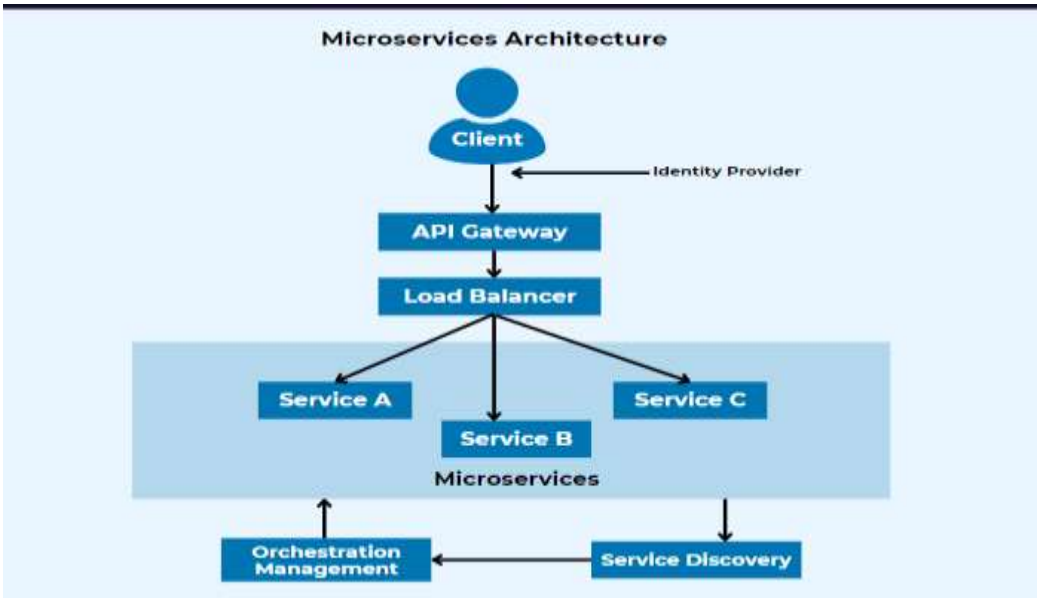

Fig 5. system architecture

6. MODULES EXPLANATION

6.1 Admin Module

- User Management
- Service Category Management
- Workflow Configuration
- Task Assignment
- Request Approval/Rejection
- Monitoring Dashboard
- Reports & Analytics

6.2 User Module

- Registration & Login
- Create Service Request
- View Request Status
- Track Workflow Progress
- Receive Notifications

6.3 Workflow Automation Module

- Automated routing
- Stage transitions
- Approval validation
- Escalation rules

6.4 Notification Module

- Email/Alert notifications
- Status updates
- Approval notifications

7. IMPLEMENTATION

The system was developed using:

- **Frontend:** HTML, CSS, JavaScript, React
- **Backend:** Java, Spring Boot, REST APIs
- **Database:** MySQL
- **Authentication:** Role-Based Login System
- **Server:** Embedded Apache Tomcat
- **Tools:** IntelliJ IDEA, VS Code, Postman

Development included:

- Secure authentication setup
- Workflow rule configuration

- CRUD operations for requests
- Dashboard analytics integration
- Testing and debugging

8. RESULTS & ADVANTAGES

8.1 Results

- Automated workflow execution
- Reduced manual intervention
- Real-time tracking of service requests
- Secure role-based access
- Improved communication
- Accurate data storage

The system successfully processes requests from submission to closure with transparency and efficiency.

8.2 Advantages

- Reduced paperwork
- Improved accountability
- Faster request processing
- Real-time notifications
- Secure data management
- Scalable architecture
- Suitable for enterprise deployment

9. CONCLUSION

The Service Request and Workflow Automation System provides a structured, automated, and centralized solution for managing service requests efficiently. By incorporating workflow automation, role-based access control, and real-time tracking the system improves operational efficiency and user satisfaction
The project demonstrates the effective use of Java Full Stack Technologies in building scalable enterprise solutions

10. FUTURE SCOPE

The system can be enhanced with:

- Mobile application integration
- Email & SMS notifications
- AI-based request prioritization
- SLA (Service Level Agreement) monitoring
- Cloud deployment
- Advanced analytics dashboard
- Integration with ERP systems

These improvements will make the system more powerful and enterprise-ready.

11. REFERENCES

1. Research Papers & Articles 1. P. P. Saha, A. Roy, and S. Banerjee (2021)
"Workflow Automation Systems for Enterprise Applications."
Discusses automation techniques for managing service requests and workflows in organizations
2.K. Srinivasan and R. Kumar (2020) "Design and Implementation of Web-Based Service Desk Systems." Explains the architecture and benefits of online service request management systems
3. M. Fowler (2018)
"Microservices Architecture and Enterprise Application Design."
Provides insights into backend service design using RESTful APIs and microservices.
4.R. Fielding (2000)
"Architectural Styles and the Design of Network-based Software Architectures."
Introduces REST architecture, which forms the basis of modern web services.

5. Spring Boot Official Documentation
https://spring.io/projects/spring-boot
Provides detailed guidance on building backend applications using Spring Boot.
6. Spring Data JPA Documentation
https://spring.io/projects/spring-data-jpa
Explains database interaction and ORM concepts used in the project
7. MySQL Official Documentation
https://dev.mysql.com/doc/
Reference for relational database design and query execution.
8. Apache Tomcat Documentation
https://tomcat.apache.org/
Used as the embedded web server for running the backend application.
9. Spring Boot REST API Examples
GitHub: https://github.com/spring-projects/spring-boot
Contains sample implementations of RESTful services.
10. Service Desk Management System – Java
GitHub: https://github.com/dprasad9398/Service-request-Workflow-Automation-System
Source code repository of the implemented project.
11. Java Full Stack Project Examples
GitHub: https://github.com/topics/spring-boot-project
Provides reference implementations for enterprise-level Java projects

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